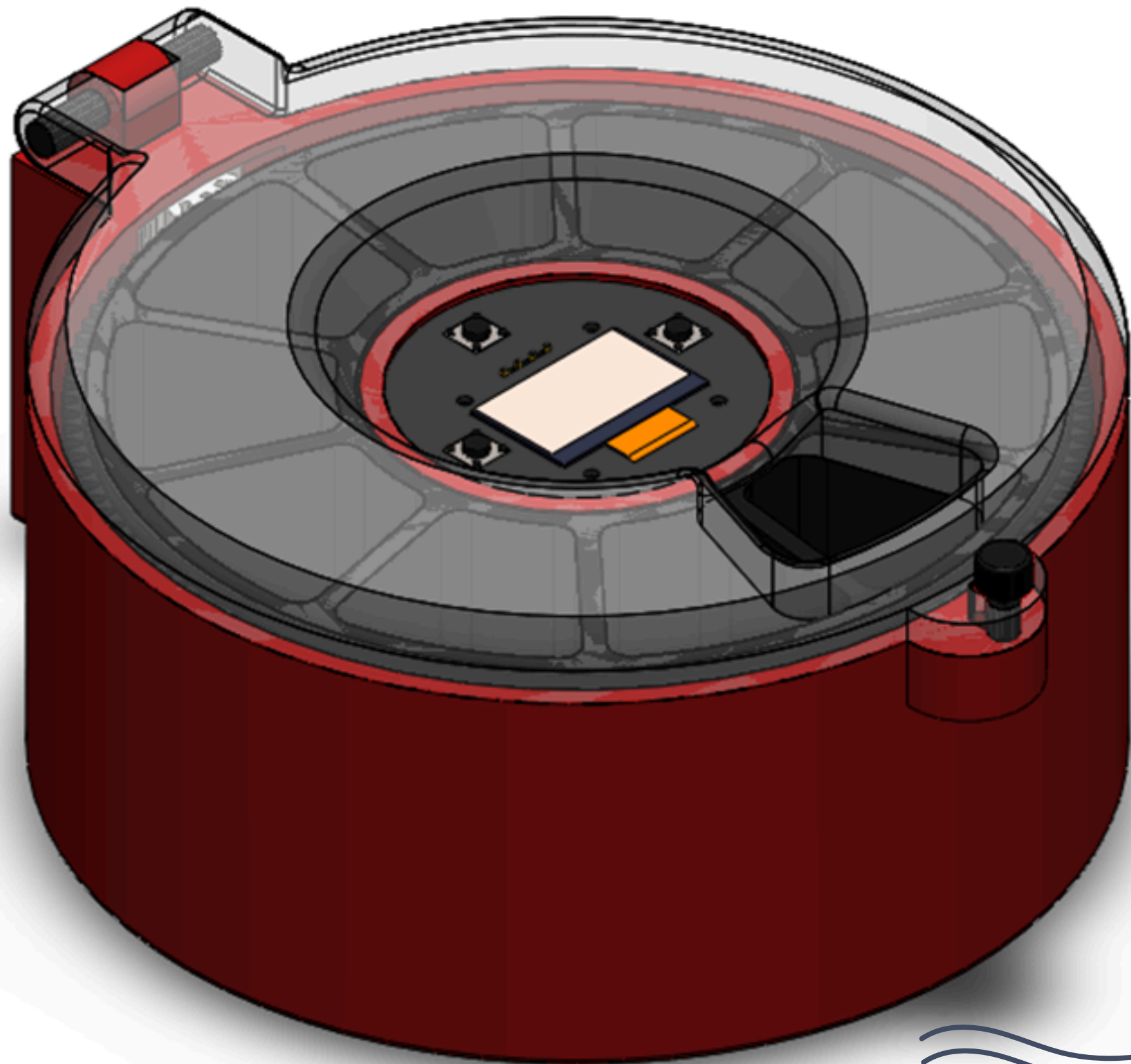


Medication dispenser



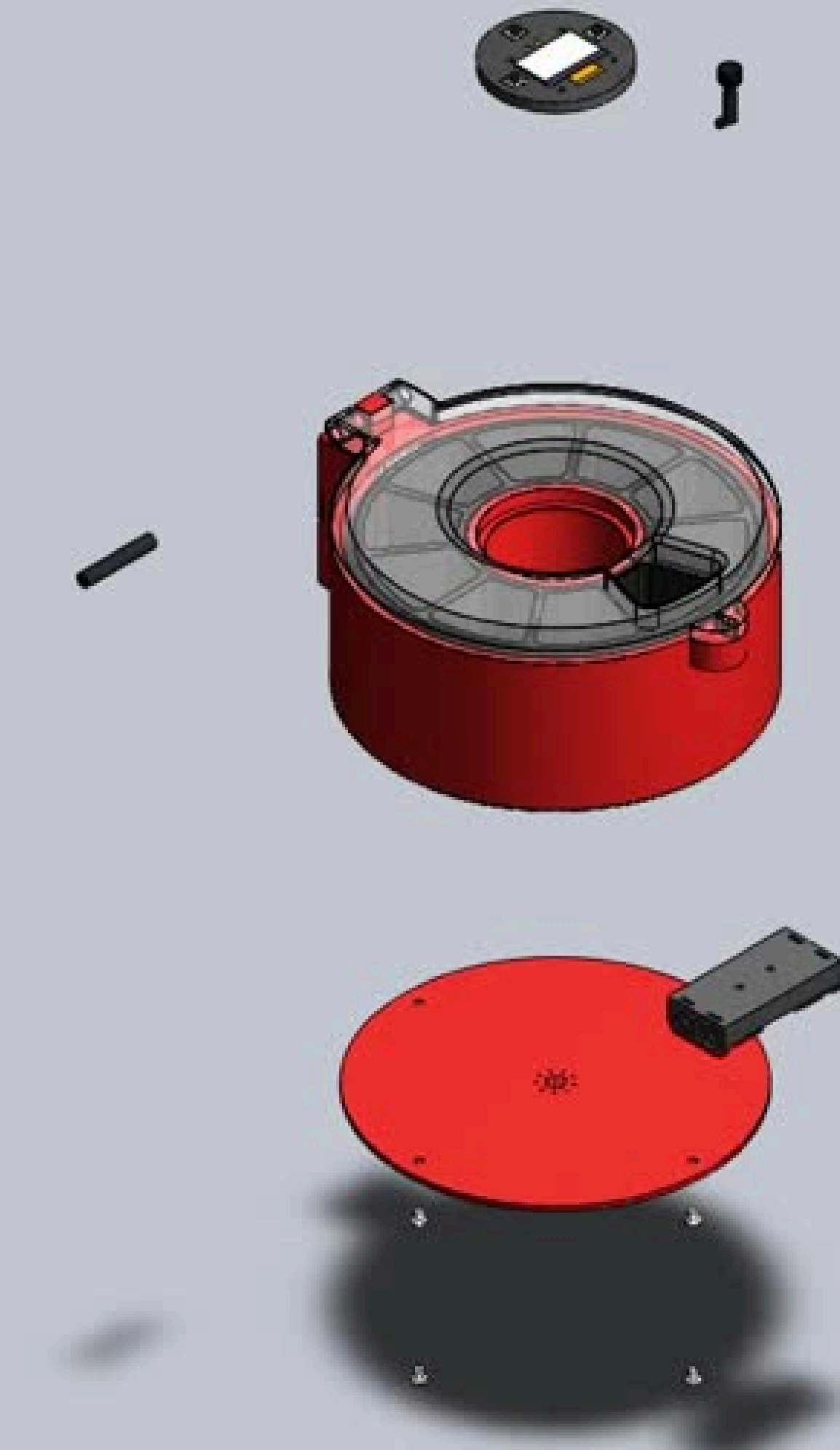
Prepared by
Ashot Nahapetyan
Lia Martirosyan

Instructors
Sargis Zeytunyan
Levon Stepanyan

American University of Armenia
ENG-241 Computer-Aided Design

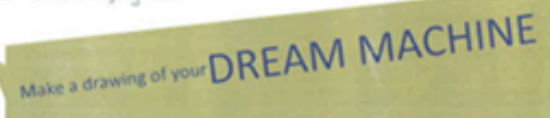
Abstract

- The Medication Dispenser machine is an automated solution designed to remind patients and the elderly about taking medication on time and optimize the treatment plan.
- It allows its users to not only take the medication on time, but also simplifies the process for searching for the right pill for patients with several cognitive disabilities.



A decorative graphic consisting of several horizontal, wavy lines in a light blue-grey color, positioned at the bottom of the page.

The inspiration and idea was suggested by a little girl named Nelli, who dreamed of having a machine that reminds her grandparents to take their medications on time.



Name of the dream machine	Neurological-Imaging by Functional MRI (fMRI) and PET scans
What is it doing?	Measuring brain activity and blood flow changes
Why did you invent it?	To understand the neural basis of thought and behavior



Problem Statement

Many individuals, especially the elderly or those with specific cases, often forget to take their doses at the correct times due to memory limitations, busy routines, or complex multi-dose prescriptions.



Objectives

- ensure proper and accurate medication delivery
- enhance user safety
- support effective treatment
- provide trends about user health statistics
- create a user-friendly interface
- suit different treatment plans
- ensure practicality
- be eco-friendly



Analysis of Existing Solutions



Pivotell Automatic Pill Dispenser Mk3

Live Fine Automatic Pill Dispenser

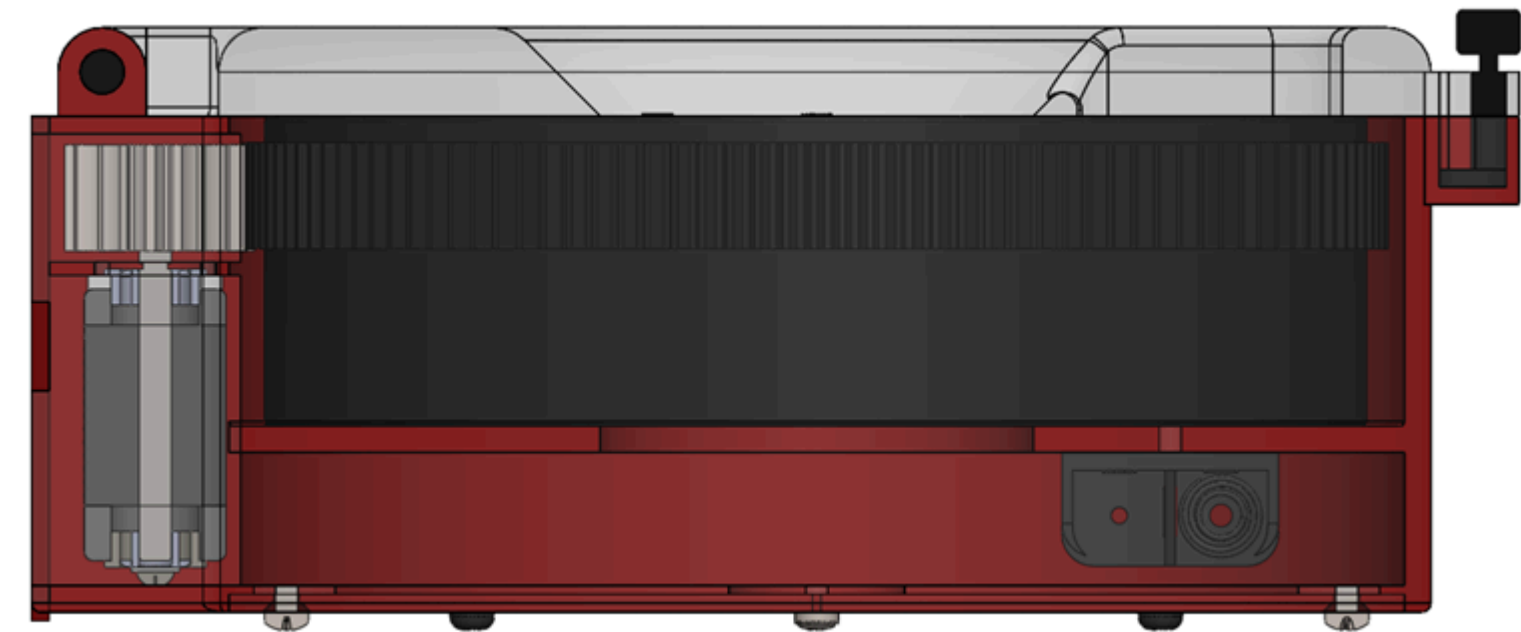
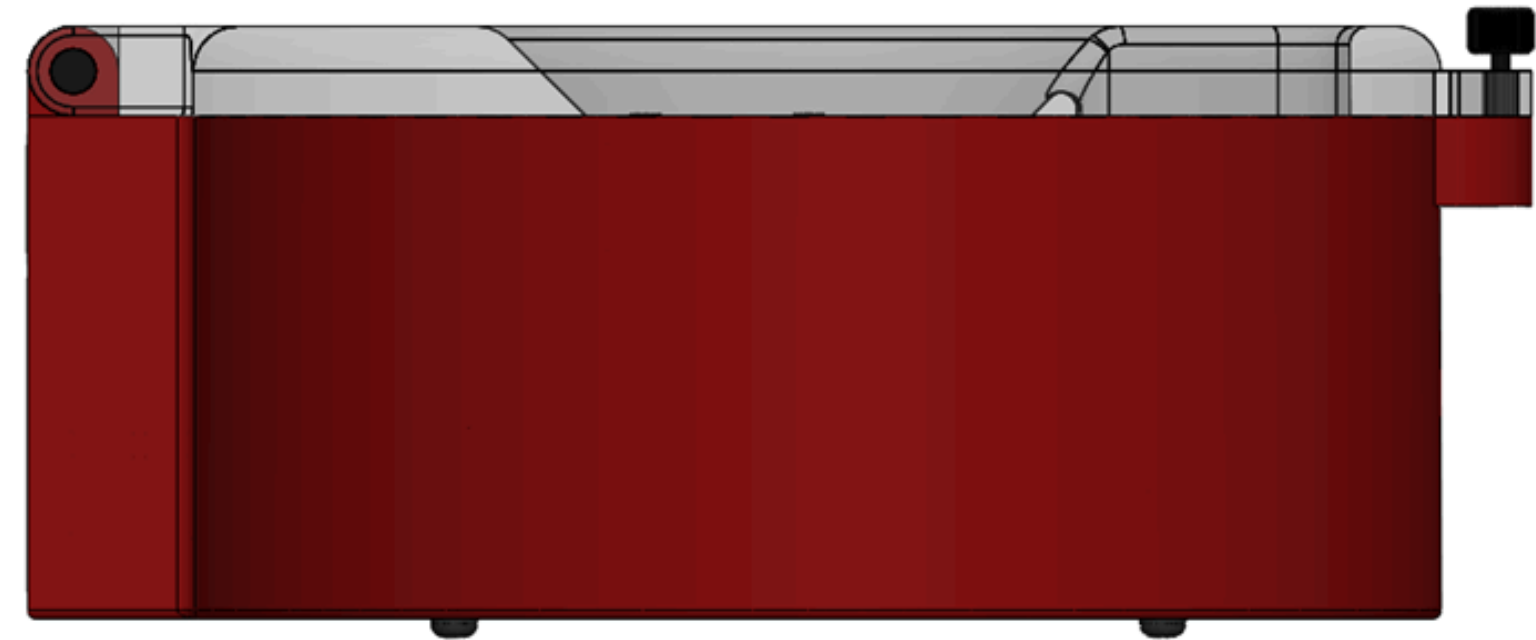
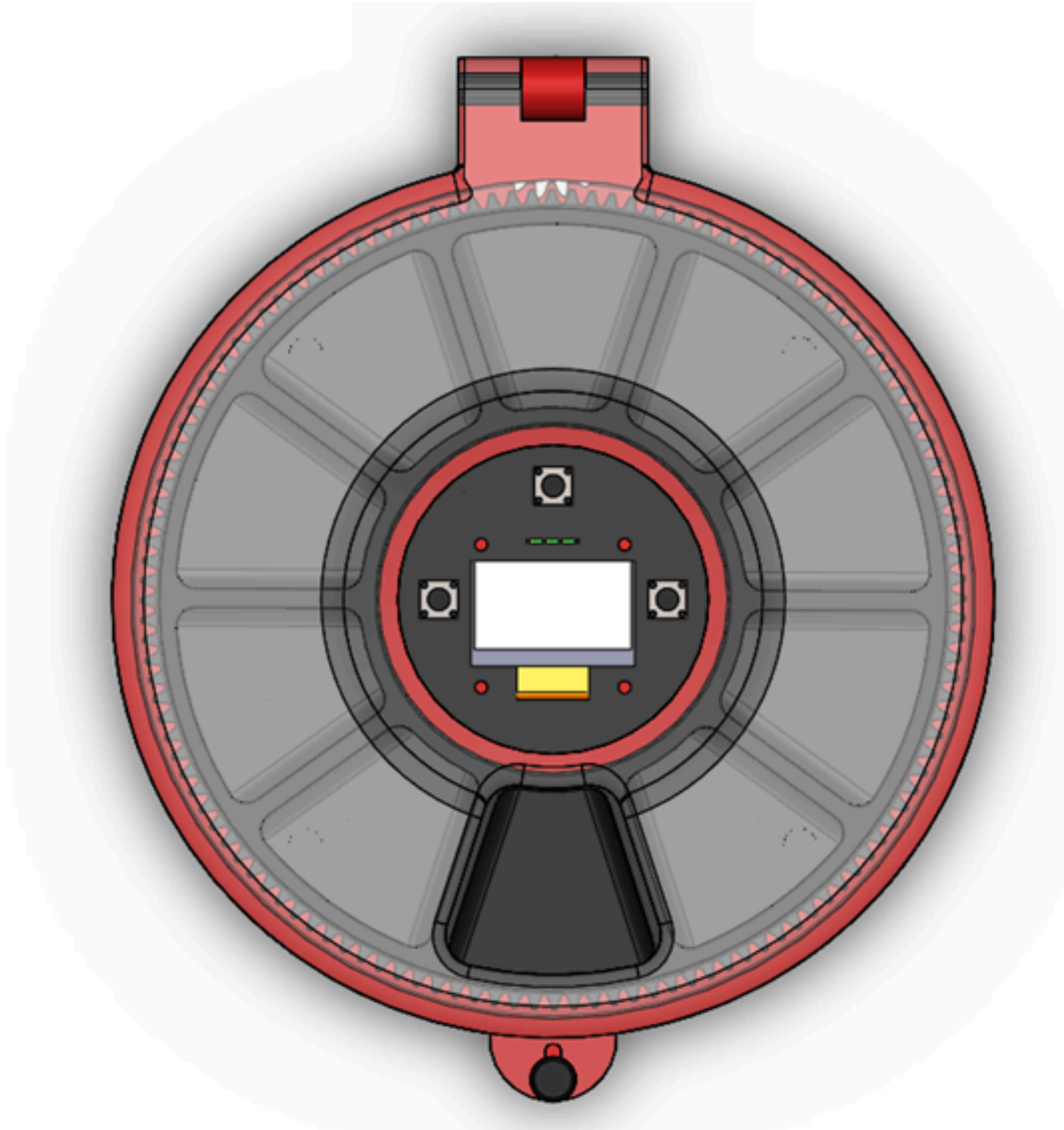


System Requirements & Constraints

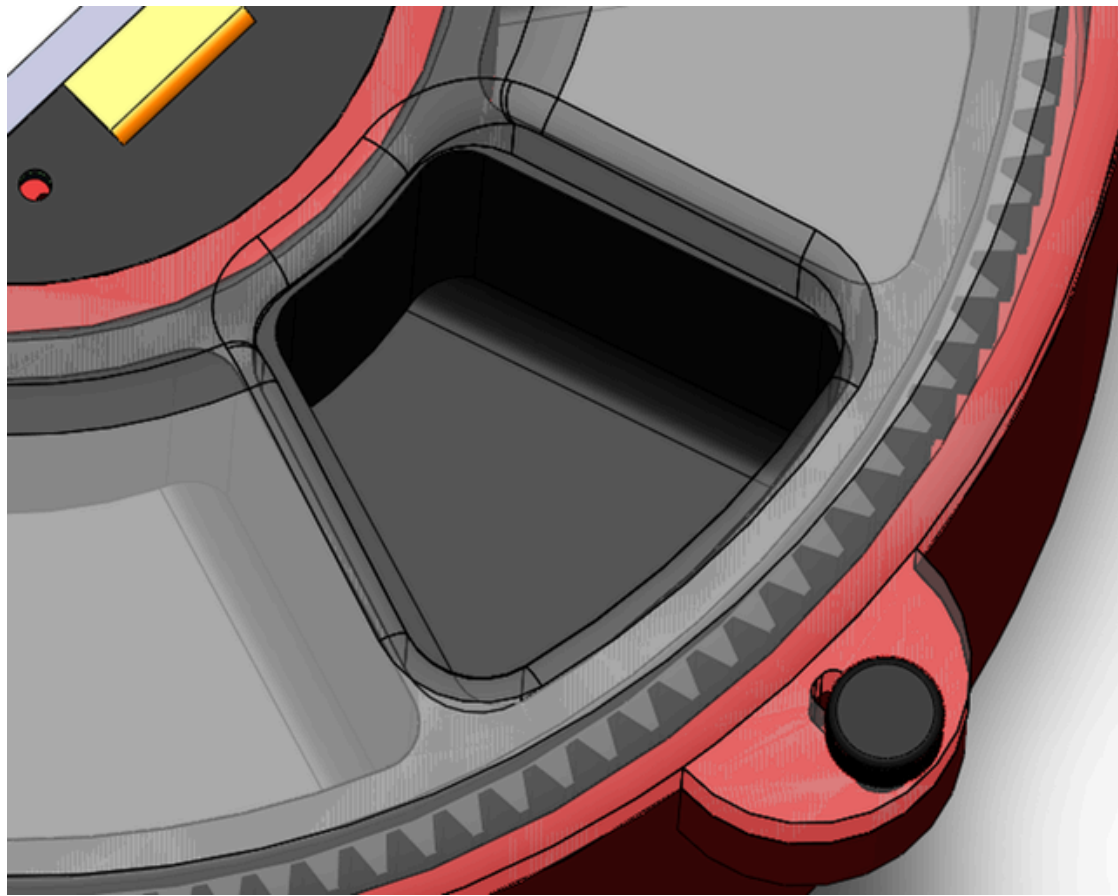
- **provide clear reminders through indexes such as light or sound**
- **prevent users from accessing multiple preset doses once**
- **be reusable and allow stakeholders to recycle it**

- **the material should be neutral and not react with the medication inside it**
- **weight and dimensions of the system**
- **affordability of the components matching our criteria**

Final Design Description

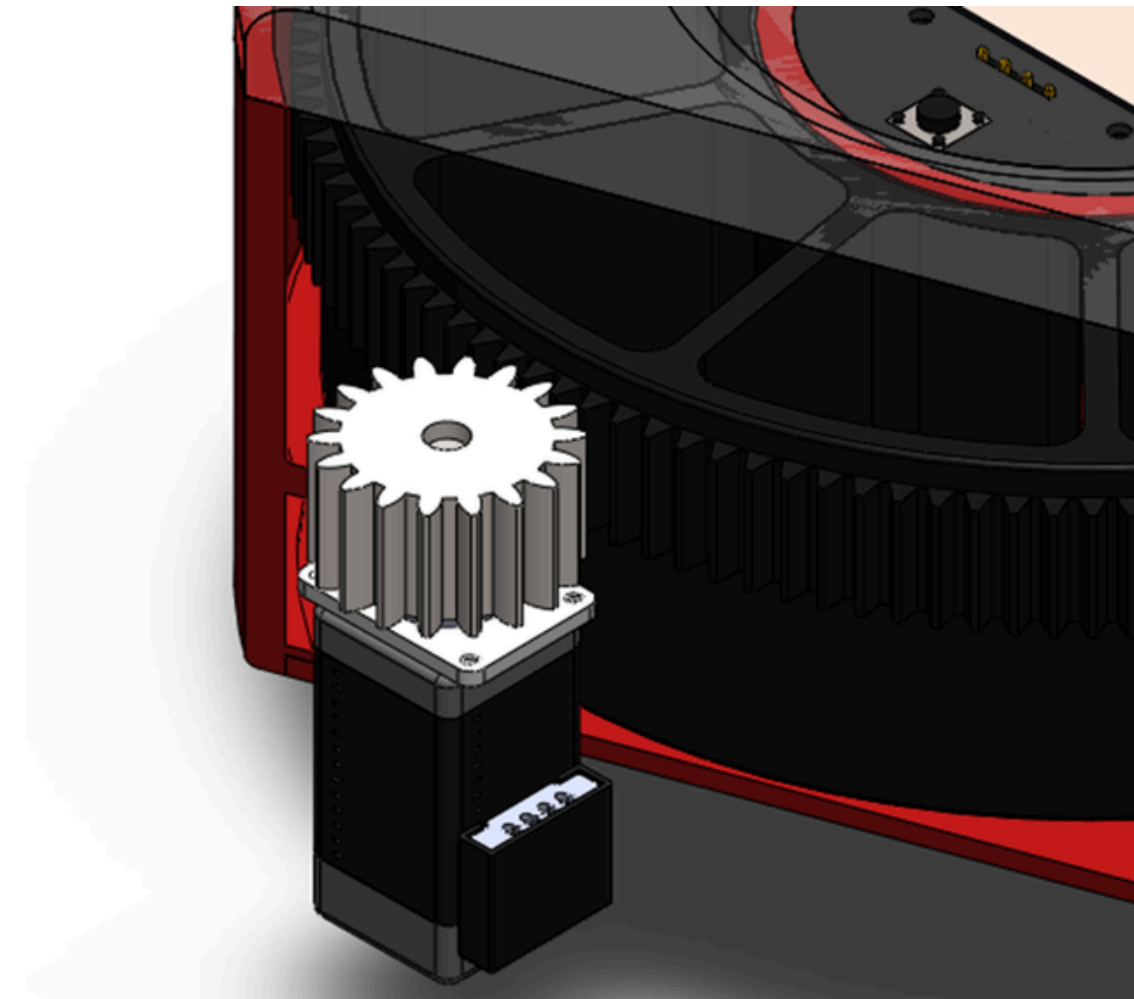


Final Design Description



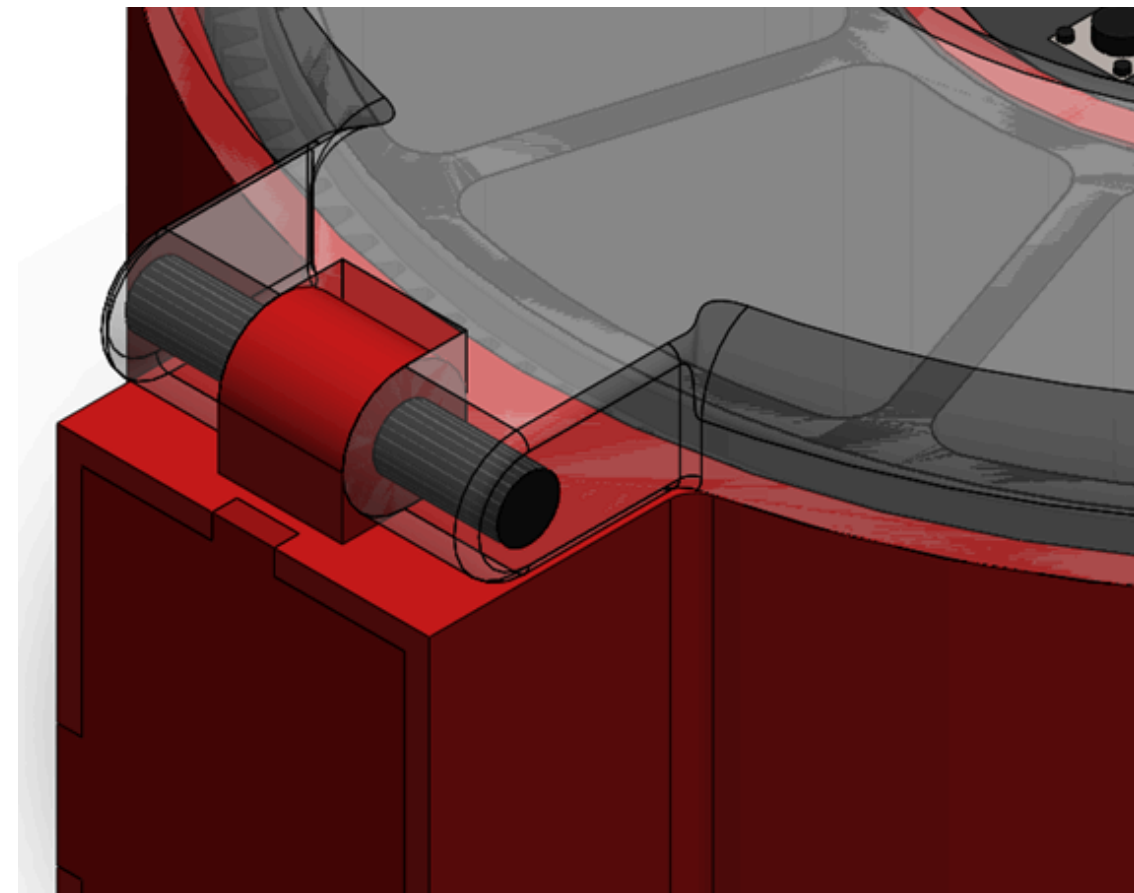
Designated Medicine Pockets

Gear view



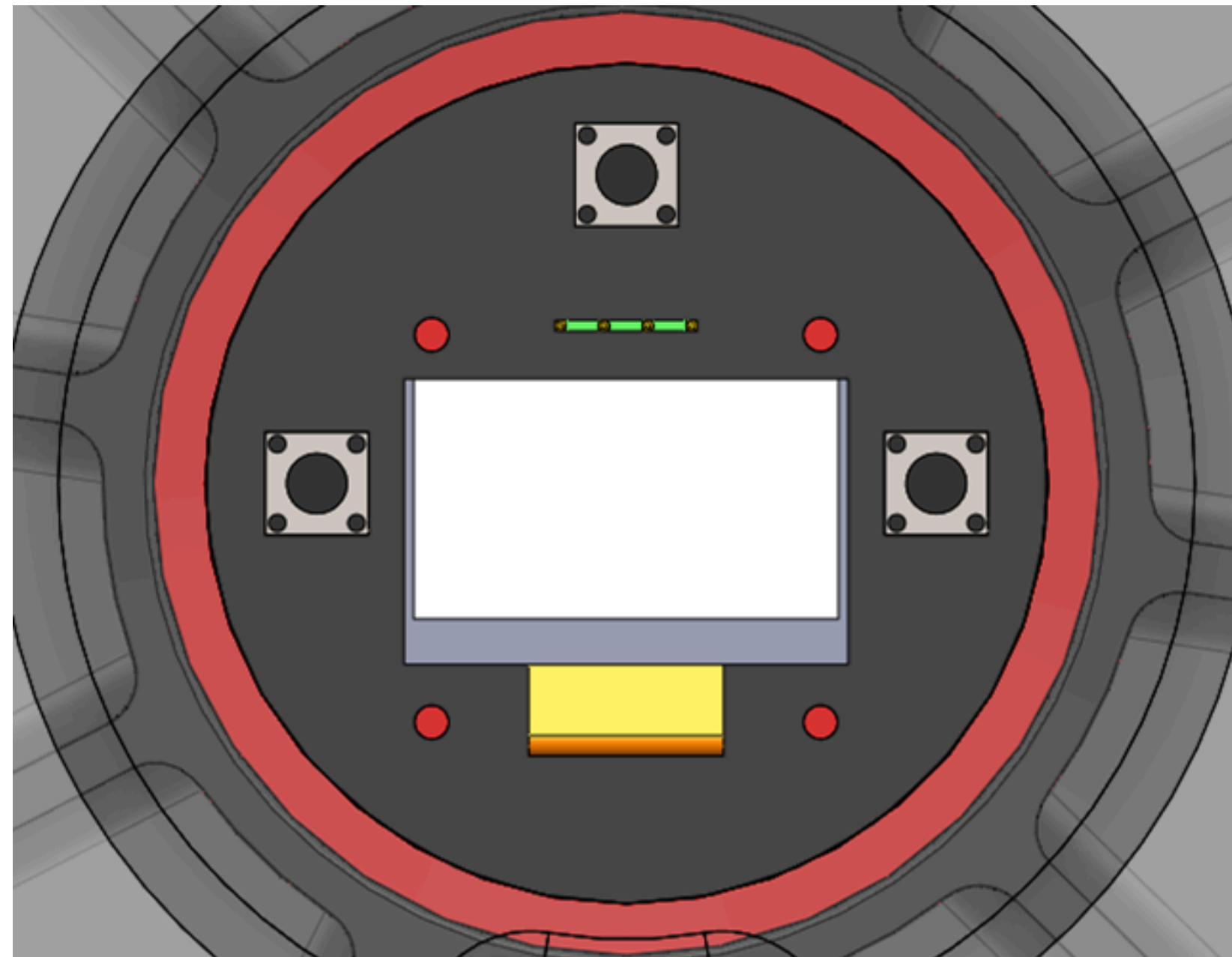
Final Design Description

Safety Lock

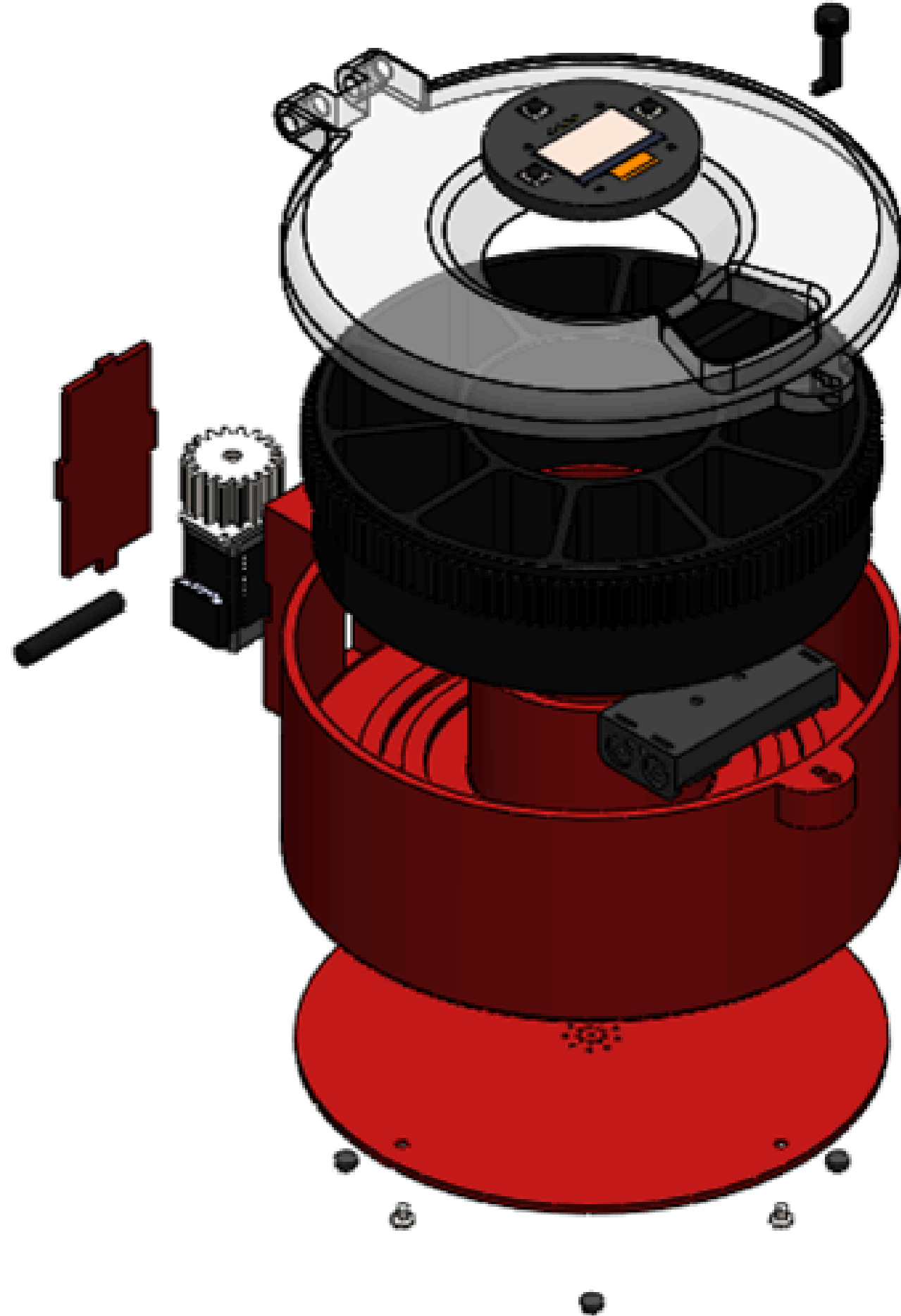


Hinged Lid Design

End-User Interface



Exploded View



Materials

BOM table:

ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	Shell		1
2	Rotary Wheel With Gear		1
3	Lid		1
4	Hinge Pin		1
5	Stepper Motor		1
6	OLED TEMU 0.96 In	PART-OLED TEMU 0.96 In-DESC	1
7	Push_Button v3		3
8	Cover		1
9	Key		1
10	Battery Case		1
11	Bottom Lid		1
12	Rubber Legs		4
13	Stepper Motor Gear		1
14	Side Cover		1
15	B18.6.7M - M2.5 x 0.45 x 3 Type I Cross Recessed PHMS -3N		4

The Materials selected for the machine include lightweight medium **gloss plastics** for the Shell, Lid, Hinge Pin, Cover, Bottom and Side covers, Battery Case. **Medium gloss Rubber** for Rubber Legs. Metal for Stepper Motor Gear, and **Polypropylene** for the Rotating Wheel.



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 5. Free CAD designs, files & 3D models | The GrabCAD Community Library. (n.d.-d).
<https://grabcad.com/library/mpd-bc2aaapc-battery-holder-1>
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Thank you!